

Network intrusion detection based on balancing generative adversarial network

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Abstract—Aiming at the network security problems faced by resource-constrained Internet of Things (IoT) edge devices, this paper proposes an intrusion detection method based on balancing generative adversarial network, namely BAGAN-BD, which uses BAGAN-BD for data enhancement to balance the data distribution, so as to solve the problem that highly unbalanced data will affect the classification detection performance. BAGAN-BD firstly inherits the prior knowledge of the autoencoder with tag embedding, then adopts a multi-generator structure to generate different types of attack data, adds a classifier structure, uses Focal Loss to calculate classification losses, and finally introduces gradient penalty technology to solve the problems of GAN training instability and mode collapse. The experimental results show that the proposed method achieves 97.54% macro average accuracy, 96.90% recall rate and 97.19% F1 score in classification tasks on UNSWNB15 dataset, all of which are higher than the comparison model.

Keywords—Intrusion detection, Internet of Things, Data enhancement, Network security

I. INTRODUCTION

The rapid development of the Internet of Things (IoT) has brought about a surge in the amount of data, and traditional cloud computing has limitations in handling large amounts of data transmission and network response, and edge computing has become a key technology to solve this problem. By deploying computing and storage resources at the network edge, edge computing reduces latency and improves data processing efficiency, but it also brings new security problems. IoT edge devices usually have limited computing and storage resources and relatively weak defense capabilities, so there is an urgent need to strengthen the network security protection of these devices^[1].

Intrusion detection system (IDS), as an active defense technology, can not only monitor and analyze network traffic, but also identify and respond to potential security threats, so as to protect network resources from being infringed^[2]. It has been deeply studied by many scholars, and intrusion detection is classified as a classification problem. In the research field of intelligent intrusion detection system, many scholars try to use machine learning and deep learning for intrusion detection. In earlier studies, reference^[3] proposed an intrusion detection framework based on the combination of support vector machine and naive Bayes feature embedding technology, and built a more efficient intrusion detection model. However, with the rapid development of the Internet of Things, network data is growing at a geometric rate, and traditional machine learning is unable to deal with large-scale and high-latitude data, and deep learning methods

provide superior detection performance with its excellent data characterization ability. For example, the model proposed in reference^[4] cleverly integrates the advantages of convolutional neural network (CNN) and bidirectional Long short-term memory network (BiLSTM), which are used to capture the spatial features and time series features of network traffic respectively, realizing the hierarchical spatial-temporal feature learning of traffic data, but facing the problem of high computational complexity. Reference^[5] integrates BiLSTM into DQN, which provides a new idea for detecting zero-day attacks and is innovative in analyzing time-tense changes in the static classification background. However, in the face of complex and rapidly evolving zero-day attacks, its detection efficiency and accuracy face challenges, and it is difficult to deal with data imbalances.

The methods of deep learning in dealing with unbalanced data are mainly divided into data level and algorithm level. At the data level, SMOTE mainly balance the data set by oversampling the minority class samples and undersampling the majority class samples, such as SMOTE algorithm^[6] and ADASYN algorithm. For example, Li^[7] proposed a method that realizes oversampling through ADASYN algorithm and builds decision tree model combined with ID3 algorithm to carry out intrusion detection. The method at the algorithm level mainly focuses on improving the training process of the model to make it pay more attention to a few class samples. For example, Ren^[8] used EQL v2(Equalization Loss v2) to increase the weight of minority classes, balance the learning attention of minority attack classes, and effectively improve the detection rate of minority classes. Although these methods are effective in dealing with unbalanced data sets, there are still limitations. Oversampling by adding a few class samples to balance the dataset may lead to overfitting, while random undersampling by randomly removing most class samples to reduce the number of samples may lose important information. In addition, SMOTE synthesizes new samples by interpolating between a few class samples and their nearest neighbors, but this method can lead to sample overlap, especially if class boundaries are not obvious.

Generative adversarial network (GAN), as a generative model, has been widely used to solve the problem of data imbalance because of its ability to generate nearly real data. Lee^[9] used the GAN model to solve the problem of data imbalance, avoiding the weaknesses that led to data loss or overfitting in previous studies. Later, researchers proposed cGAN (conditional GAN) to generate specific categories of network attack samples to enhance the model's ability to identify specific categories of data and improve the model's generalization ability^[10]. However, the problem that GAN

cannot accurately learn the distribution of minority samples is not taken into account because the number of minority samples is too small. BAGAN(balancing generative adversarial network) proposed a method to generate images of a few classes to balance the original data set^[11], but the problem of GAN pattern collapse is still unavoidable, and the quality and classification accuracy of BAGAN's generated data need to be improved, especially when the data of different classes have similar characteristics, the generated results are unstable. Cui^[12] alleviated the problems of mode collapse and training instability in GAN training by using improved WGAN-GP (Wasserstein GAN with Gradient Penalty), and used Chi-square test to select data features, but did not provide a scheme with high detection accuracy.

On this basis, a new intrusion detection method BAGAN-BD (balancing generative adversarial network with Balanced Data) is proposed for the generation of minority samples. To reduce the impact of unbalanced data on classification detection, the main innovations are as follows:

- 1) In the autoencoder part, tag embedding tag mechanism is introduced to enhance the model's learning of category features and provide better prior knowledge for GAN.
- 2) GAN adopts multi-generator structure, and integrates classifier in GAN to calculate classification loss using focus loss, so that the model can focus more on a few class samples, and effectively generate sample data with high quality and low overlap between classes.
- 3) In order to solve the problem of instability and model collapse in GAN training, a gradient penalty mechanism is introduced to help stabilize the training process and ensure the diversity and quality of generated samples.

II. METHODOLOGY

A. Improvement of autoencoder

BAGAN's main innovation is the use of autoencoder for GAN initialization. BAGAN uses a typical autoencoder, where the encoder converts the raw image data into a latent vector, and the decoder converts the latent vector into a reconstructed image, training the autoencoder to learn all images unsupervised by minimizing the loss between the original image and the reconstructed image. However, in this case, the potential representations of various types of different samples learned by the autoencoder will overlap, and will skew to most classes, so that it is difficult to learn a better prior distribution. In order to get a better potential representation of low overlap between classes, BAGAN-BD introduces a label embedding mechanism in the autoencoder part. Category labels are embedded in vectors with the same size as the potential vector, and the two are combined to get a new potential vector, which helps the decoder to generate a reconstructed image with more class features. The improved autoencoder structure is shown in Figure 1 and mainly consists of encoder, decoder and tag embedding module. The input data goes through the encoder to get the latent vector latent, and the label information y is embedded and fused into the latent vector to get y_latent, which then generates the reconstructed data gen_data through the decoder.

Through this improvement, the autoencoder can more effectively capture the feature structure of various types of data, which helps the subsequent generator to pre-train the weights with the autoencoder, thereby improving the generation quality of a few categories of images.

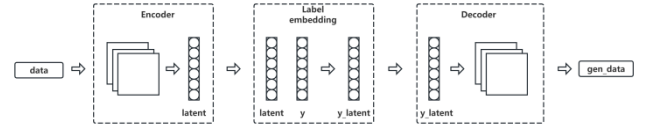


Fig. 1 Autoencoder structure diagram

B. BAGAN structure improvement

The BAGAN's discriminator has only one output, which not only determines the authenticity of the image, but also classifies the image, and while this multi-tasking design has certain advantages, it also brings performance challenges. Especially on unbalanced data sets, discriminators may not be able to take care of both tasks, resulting in poor classification performance, especially for a few categories.

Therefore, BAGAN-BD separates the task of judging the truth and the classification of categories, adding the classifier, discriminator and classifier will inherit some of the weight of the encoder, helping them to learn the potential knowledge of the label embedded in the data. And the introduction of a multi-generator structure, each generator is responsible for generating a specific class of images, which can ensure that each class of images can be fully learned and generated. These generators all get decoder weights as initialization states, and subsequent training only focuses on specific classes of images. In the sample generation phase, each generator generates an image of the corresponding category according to the category label.

Figure 2 is the BAGAN-BD structure diagram. The noise z and category label y are input into the generator, and the generator of the corresponding category will embed the two and obtain the generated sample gen_data. Discriminator D and classifier C extract the image features of the sample, and output the authenticity and classification.

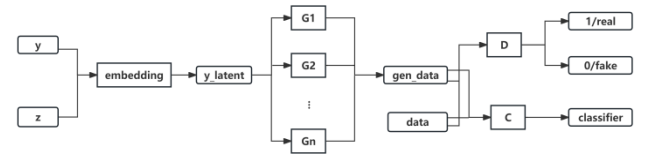


Fig. 2 BAGAN-BD structure diagram

C. Objective function improvement

The GAN model is optimized through adversarial training, where the goal of generator G is to improve the similarity of the distribution of the generated data and the goal of discriminator D is to accurately distinguish the generated data from the original data. GAN train G and D using the objective function of equation(1) :

$$\min_G \max_D L(D, G) = E_{x \sim p_r(x)} [\log(D(x))] - E_{z \sim p_z(z)} [\log(D(G(z)))] \quad (1)$$

Where P_r is the distribution of real data, P_z is the distribution of generated data, x represents the original traffic data, z is the noise sampled from the Gaussian distribution, $z \sim P(z)$, $x^* = G(z)$ represents the sample data generated by the generator.

BAGAN-BD still follows the adversarial training of G and D, but different from GAN, each generator in the model in this paper only needs to focus on the generation of class i

samples, and uses classifier to identify the category of input samples, and uses classification loss to update generator and discriminator D, the target loss is shown as follows:

$$\min_G \max_D L(D, G) = \sum_{i=1}^N [E_{x \sim p_r(x)} [\log(D(x))] - E_{z \sim p_z(z)} [\log(D(G_i(z)))] - E_{x \sim p_r(x)} [FL(x)] + E_{z \sim p_z(z)} [FL(z)]] + GP \quad (2)$$

Where $FL(x)$ represents the classification loss of the classifier to the original data x , which is used to update the discriminator D , $FL(z)$ represents the classification loss of the update generator, and GP represents the gradient penalty term introduced to alleviate the instability problem during BAPAN training.

BAPAN uses the standard cross-entropy loss function to deal with class losses. However, when the data set is unbalanced, the cross-entropy loss pays more attention to the majority class samples, and it is easy to ignore the minority class samples, which makes it difficult for the generator to generate high-quality minority class data. Therefore, Focal Loss was introduced to calculate the classification loss. Focal Loss adjusts the loss of easily classified samples by adjusting factor γ , making the model pay more attention to the difficult classified samples, helping the generator to better learn the distribution of samples of a few classes and improve the quality of sample generation. The specific formula for focal loss is as follows:

$$\text{Focal Loss} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (3)$$

Where α_t is the class weight, used to adjust the influence of different classes, p_t is the prediction probability of the model for the correct class, and γ is the regulator, used to adjust the weight of the sample.

In order to solve the problem of instability and mode collapse during BAPAN training, BAPAN-BD introduces Gradient Penalty into the loss function. Gradient Penalty is a regularization technique used in WGAN-GP (Wasserstein GAN with Gradient Penalty)^[13]. By restraining the gradient of the discriminator, the confrontation between the generator and the discriminator is balanced, so that the generator can learn the distribution of real data more easily. The specific formula of the added gradient penalty term is as follows:

$$GP = \lambda E_{x^* \sim p(x^*)} [(\|\nabla_{x^*} D(x^*)\|_2 - 1)^2] \quad (4)$$

Where $x^* = \alpha x_r + (1 - \alpha x_f)$ represents the interpolated sample of real sample x_r and generated sample x_f , and λ is the gradient penalty factor.

III. RESULTS

A. Experimental data set selection

This experiment was run under Windows operating system, using Python programming language and PyTorch framework to build the model, the CPU was Intel Core i5-13600KF, the memory was 32 GB, and the GPU was NVIDIA GeForce RTX 4070 SUPER.

The UNSW-NB15 dataset was selected for simulation experiments. The UNSW-NB15 dataset is a network intrusion detection dataset provided by the Cyber Security Lab at the University of New South Wales in Sydney. This

dataset simulates network traffic in a real network environment and includes a variety of common network attacks and normal traffic. The UNSW-NB15 dataset contains 175,341 network connection records, each of which contains 49 characteristics, including three character characteristics and a label, including profile information, network connection characteristics and traffic statistics, covering all aspects of network connection. Network connections in the dataset were flagged as either normal traffic or nine other different types of attacks, such as Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms. The detailed description and classification of attack types are provided. The UNSW-NB15 dataset contains a mixture of real modern normal activity and synthetic contemporary attack activity, which makes it ideal for evaluating the performance of modern network intrusion detection systems^[2].

In this study, 5 types of data are selected for experiments, including normal data and 4 types of attack data. These four types of attacks are the most common and harmful attacks, and can be classified into a few types in quantity. Training data are shown in Table I.

TABLE I. UNSW-NB15 DATASET INFORMATION

Label	Class	Quantity	IR ^a
0	Normal	56000	-
1	Dos	10491	5.38
2	Backdoors	1746	32.07
3	Shellcode	1133	49.43
4	Worms	130	430.77

^a IR represents the ratio of normal samples to attack samples.

B. Data preprocessing

1) Data cleaning

Firstly, the CSV file of the UNSWNB15 dataset was preprocessed, and the missing values and infinite data items were deleted. For duplicate data, the first record was retained and the remaining duplicate records were deleted. In addition to columns like id and data that are not strongly related, the remaining features include three character features, proto, state and service. The protocol type of proto involves a wide range of 133 types, and the traffic data of all protocol types may be attacked, which is not directly related to whether the network traffic is benign, so it is discarded. The remaining two character type features are state and service, in which state feature has 9 symbolic attributes and service feature has 13 symbolic attributes, and these two character type features are uniquely coded.

2) Data normalization

After digital coding, it is necessary to normalize the data to standardize its distribution. There are multiple attributes in the data set, and the value size and range of each attribute are different. In order to keep all indicators in the same order of magnitude, this study uses the Min-Max normalization algorithm to process the data set, and the calculation formula is shown as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (5)$$

Where x represents the initial feature value, $x_{\min}$ represents the minimum feature value in the attribute, $x_{\max}$ represents the maximum feature value in the attribute, and x' represents the value after normalization. Each eigenvalue is

transformed to fit the range [0,1]. The normalization of the data can improve the convergence speed and accuracy of the model, and also can prevent the gradient explosion problem.

C. Evaluation indicators

Accuracy rate (P), recall rate (R) and F1 score (F1) are used as evaluation indicators. The indicators of category i are shown as follows:

$$P_i = \frac{TP_i}{TP_i + FP_i} \quad (6)$$

$$R_i = \frac{TP_i}{TP_i + FN_i} \quad (7)$$

$$F1_i = \frac{2 \times P_i \times R_i}{P_i + R_i} \quad (8)$$

Where TP (True Positives) is the number of true cases, that is, the number of instances that the model correctly predicts to be positive; FN (False Negatives) is the number of false negatives, i.e. the number of positive instances of a negative class that the model incorrectly predicts. FP (False Positives) represents the number of false positives, i.e. the number of negative class instances that the model incorrectly predicts as positive. The accuracy rate measures the proportion of categories that are actually positive out of all categories that are actually positive, and the recall rate measures the proportion of all categories that are actually

positive that are correctly predicted by the model to be positive.

The macro average accuracy rate, recall rate and F1 score are obtained by calculating the corresponding indicators of each category and averaging, regardless of the sample number of each category, all categories are treated equally. In multi-class classification tasks, especially when there is a large difference in the number of samples between classes, the macro average index can balance the influence of different classes on the overall evaluation results, and ensure that the model has a good performance on all classes.

D. Comparative analysis of experimental results

In order to analyze the improvement of the detection performance of the improved BAGAN model in terms of data enhancement, this section compares the overall classification performance after training between the proposed model and there other data-enhanced mixed datasets. ADASYN, WGAN-GP and BAGAN are selected for comparative experiments. EfficientNetB0 was selected for all classification models, and EfficientNet was one of the models that achieved the best accuracy in ImageNet image classification tasks. A few classes were selected for each data enhancement scheme to generate 10,000 data, and too many samples were easy to cause redundancy and noise.

TABLE II. COMPARISON OF EFFECTS OF DATA ENHANCEMENT SCHEMES

	ADASYN			BAGAN			WGAN-GP			BAGAN-BD		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
0	99.52%	99.43%	99.47%	95.54%	99.23%	97.35%	99.04%	99.23%	99.13%	99.56%	99.49%	99.52%
1	88.13%	93.16%	90.57%	88.43%	90.05%	94.19%	86.82%	95.76%	91.07%	93.50%	97.76%	95.58%
2	52.47%	34.11%	41.34%	94.48%	84.10%	88.99%	93.21%	81.27%	86.83%	96.75%	90.82%	93.15%
3	82.47%	74.42%	78.24%	98.72%	90.05%	94.19%	98.52%	86.50%	92.12%	98.24%	97.85%	98.05%
4	99.47%	100.0%	99.87%	96.71%	93.49%	95.07%	100.00%	97.78%	98.88%	99.66%	99.62%	99.64%
Macro	84.46%	80.22%	81.90%	94.77%	91.42%	92.98%	95.52%	92.11%	93.61%	97.54%	99.62%	97.19%

As can be seen from the Table II, in the UNSW-NB15 dataset, the model proposed in this paper achieved optimal results in macro average accuracy, recall rate and F1 score, with accuracy rate, recall rate and F1 value being 97.54%, 96.90% and 97.19%, respectively. ADASYN's performance improved significantly in a few categories, but this improvement did not work in all the categories where it did not work well. BAGAN saw significant improvements in Categories 2 and 3, but saw a drop in recalls in categories 2 and 4. WGAN-GP combines category weights and gradient penalties, improving performance on Category 4 but slightly decreasing accuracy on other categories. The model

presented in this paper achieves the highest accuracy and F1 scores in all categories, with the most significant improvement in category 1 and Category 2. This shows that the proposed method solves the class imbalance problem effectively and improves the overall performance of the model.

E. Ablation experiment

Finally, ablation experiments were used to verify the effectiveness of each improvement in the model in this paper. The experimental results are shown in Table III.

TABLE III. COMPARISON OF EFFECTS OF DATA ENHANCEMENT SCHEMES

	Multi-generator	gradient penalty	Focal Loss		0	1	2	3	4	Macro
				Precision	99.98%	80.31%	80.69%	63.94%	53.20%	75.42%
1	×	×	×	Recall	99.08%	88.14%	59.53%	66.77%	56.08%	73.92%
				F1	99.03%	84.04%	68.51%	65.32%	54.60%	74.30%
				Precision	99.23%	84.73%	98.63%	99.64%	99.95%	96.43%
2	✓	×	×	Recall	98.61%	97.94%	84.78%	90.11%	98.74%	94.03%
				F1	98.92%	90.86%	91.18%	94.63%	99.36%	94.99%
				Precision	98.92%	87.71%	98.81%	99.81%	99.99%	97.05%
3	✓	✓	×	Recall	99.14%	97.53%	86.88%	91.84%	98.58%	94.80%
				F1	99.03%	92.36%	92.46%	95.66%	99.29%	95.76%
				Precision	99.43%	93.43%	95.09%	98.44%	99.80%	97.24%
4	✓	✓	✓	Recall	99.46%	96.26%	91.20%	97.31%	99.45%	96.74%
				F1	99.45%	94.82%	93.10%	97.87%	99.63%	96.97%

Group 1 models performed well in the main category (category 0), but had lower accuracy, recall, and F1 scores in

a few other categories. After the introduction of multiple generators in group 2, the model's accuracy, recall, and F1

scores improved across all categories, especially in a few categories. This shows that multiple generators help to improve the quality of model generation for a few classes. The addition of gradient penalties to the multiple generators in group 3 further improves model performance, especially in terms of accuracy and F1 scores. Gradient punishment helps to stabilize the training process of GAN and reduce the mode collapse. The introduction of Focal Loss in Group 4 further improved the recall rate and F1 score of the model in a few categories. The use of Focal Loss made the model pay more attention to samples that were difficult to classify, thus improving the overall performance of the model.

In summary, the ablation experiments show that the introduction of multiple generators, gradient penalty and Focal Loss all have a positive effect on improving the performance of the model when processing unbalanced data sets. The combined use of these components significantly improves the model's macro average accuracy, recall rate, and F1 score, demonstrating their effectiveness in improving the performance of intrusion detection systems.

IV. CONCLUSIONS

In order to deal with the vulnerability of Internet of Things (IoT) edge devices to network attacks, an intrusion detection model BAGAN-BD with improved balanced generation adversarial networks is proposed. BAGAN-BD was adopted to learn the distribution of original data, and gradient penalty technology was used to effectively solve the problem of pattern collapse. In order to improve the quality of the generated samples, a multi-generator structure was adopted to generate data from a few classes. A classifier-assisted generator was introduced to generate data, and Focal Loss was used to calculate class loss, thereby increasing the focus on difficult-to-classify samples. The comparative experiment results based on the UNSW-NB15 dataset show that the intrusion detection model proposed in this study can effectively identify all kinds of attacks, especially the attacks of a few categories of samples, and its detection accuracy is higher than that of the comparison method. Finally, we verify the effectiveness of the

improvement of each part of the model through ablation experiments.

REFERENCES

- [1] Ma Zheng, Yuan Lingyu, Yu Yongtao. Dynamic Weight-Based Intrusion Detection Model for IoT Integration[C]. 2023 5th International Conference on Artificial Intelligence and Computer Applications, Dalian, China, 2023: 286-290. DOI: 10.1109/ICAICA58456.2023.10405514.
- [2] Deng Miaolei, Kan Yupei, Sun Chuanchuan. Summary of network intrusion detection systems based on deep learning[J]. Journal of Computer Applications(in Chinese),2024:1-13.
- [3] Yao Wei, Shi Han, Zhao Hai. Scalable anomaly-based intrusion detection for secure Internet of Things using generative adversarial networks in fog environment[J]. Journal of Network and Computer Applications, 2023, 214.
- [4] Wang Wei. Deep Learning for Network Traffic Classification and Anomaly Detection[D]. University of Science and Technology of China, 2018.
- [5] Wu Yali, Hu Yanghu, Wang Junhu. An active learning framework using deep Q-network for zero-day attack detection[J]. Computers & Security, 2024, 139: 103713.DOI:10.1016/j.cose.2024.103713.
- [6] Seo J H, Kim Y H. Machine-Learning Approach to Optimize SMOTE Ratio in Class Imbalance Dataset for Intrusion Detection[J]. Computational intelligence and neuroscience, 2018, 20189704672.
- [7] Li Yue, Xu Wushen, Li Wei. Research on hybrid intrusion detection method based on the ADASYN and ID3 algorithms[J]. Mathematical biosciences and engineering : MBE,2022,19(2):2030-2042.
- [8] Ren K Y, Yuan S, Zhang C. CANET: A hierarchical CNN-Attention model for Network Intrusion Detection[J]. Computer Communications, 2023:170-181.
- [9] Lee J, Park K. GAN-based imbalanced data intrusion detection system[J]. Personal and Ubiquitous Computing, 2019, 25(1): 1-8.
- [10] Zhao Guangyu, Liu Peng, Sun Ke. Research on data imbalance in intrusion detection using CGAN[J]. PloS one, 2023, 18(10): e0291750. DOI:10.1371/journal.pone.0291750.
- [11] Mariani G, Scheidegger F, Istrate R. Bagan: Data augmentation with balancing gan [Online]. Available:arXiv:1803.09655.
- [12] Cui Zicai, Zhong Bocheng, Zhao Xinyang. Intrusion detection data augmentation based on GAN and feature selection technique[J]. Intelligent Computer and Applications(in Chinese), 2024, 14 (03): 174-180.
- [13] Huang Gaofeng, Jafari A H. Enhanced balancing GAN: minority-class image generation[J]. Neural Computing and Applications, 2021, 35 (7): 1-10